

# AAC Consumer Edition - Raspberry Pi Pico 2 Pinout

REVISION A | INTERNAL / PRIVATE - DIMENSION EIGHT LABS

| Device / Driver        | Pico 2 GPIO | Physical Pin | Signal / Connection | Notes                                                                |
|------------------------|-------------|--------------|---------------------|----------------------------------------------------------------------|
| NEO-7 GPS              | GP0         | 1            | UART0 TX -> GPS RX  | Production GPS; 9600 baud, 8N1                                       |
| NEO-7 GPS              | GP1         | 2            | UART0 RX <- GPS TX  | Production GPS; 9600 baud, 8N1                                       |
| NEO-7 GPS              | GP2         | 4            | PPS input           | Production GPS; rising-edge PPS input                                |
| STDOUT display         | -           | -            | USB CDC             | Default display when none is explicitly selected                     |
| VFD display (optional) | GP4         | 6            | UART1 TX only       | Allocated only when VFD driver is selected                           |
| WS2812 8x32 Panel      | GP8         | 11           | DATA -> SN74HCT541N | Allocated when WS2812 driver is selected; driver not yet implemented |

## Interface Grouping

| GPIO / Interface | Interface        | Device / Driver        | Purpose                                    |
|------------------|------------------|------------------------|--------------------------------------------|
| GP0-GP2          | UART0 + PPS      | NEO-7 GPS              | Authoritative GPS time input in production |
| USB CDC          | USB serial       | STDOUT display         | Default/fallback display output            |
| GP4              | UART1 TX         | VFD display (optional) | Transmit-only serial display driver        |
| GP8              | PIO-capable GPIO | WS2812 8x32 Panel      | RGB panel data; selected driver only       |

## Bench Wiring Notes - Read Before Connecting

- **Display selection:** Display drivers are selected at compile time. If no display is explicitly selected, STDOUT is selected automatically. STDOUT is not implicitly added when a physical display driver is selected.
- **WS2812 8x32 Panel:** GP8 / physical pin 11 is the approved data assignment. The detailed SN74HCT541N level-shifter, resistor, panel wiring, power, and pixel-order information belongs in the WS2812 driver hardware document. The driver is not yet implemented.
- **VFD display:** The optional VFD driver claims GP4 / physical pin 6 as UART1 TX only. There is no VFD RX GPIO assignment. MAX3232 module wiring is device-specific and belongs in the VFD driver hardware documentation.
- **GPS:** GP0, GP1, and GP2 are physical GPS/PPS hardware only in the production profile. The development profile uses the simulated GPS/PPS source and does not initialize those receiver pins.
- **Power:** Pico 2 power entry, regulator wiring, grounds, and display power distribution are intentionally documented on separate power/interface bench sheets.

**Status:** Revision A records the current AAC-CE signal-pin contract. GPS assignments are implemented. Display-driver pins are conditional on compile-time selection; WS2812 GP8 is approved but remains uninitialized until the driver is implemented.